

GOOGLE CYBERSECURITY CERTIFICATE

tcpdump Labs

Google Cybersecurity Professional Certificate · Study Guide

Reading a raw tcpdump traffic log to trace a malware-delivery incident from its first DNS lookup through the HTTP download and the subsequent redirect to a spoofed lookalike domain.

Track	Google Cybersecurity Professional Certificate
Course	Detection and Response
Tool	tcpdump (command-line packet analyzer)
Analyst	Chayan Panchal

1 Reading the tcpdump Log

Several customers reported that visiting `yummyrecipesforme.com` prompted a file download, after which their computers slowed down. The site owner also found themselves locked out of the admin account. The raw tcpdump log for the incident reconstructs exactly what happened over the wire.



1. DNS resolution — `yummyrecipesforme.com` resolves to `203.0.113.22`

```
14:18:32.192571 IP your.machine.52444 > dns.google.domain: 35084+ A? yummyrecipesforme.com.  
14:18:32.204388 IP dns.google.domain > your.machine.52444: 35084 1/0/0 A 203.0.113.22
```



2. TCP handshake ([S] -> [S.] -> [.]) then the HTTP GET request — the download

```
14:18:36.786501 IP your.machine.36086 > yummyrecipesforme.com.http: Flags [S] ...  
14:18:36.786517 IP yummyrecipesforme.com.http > your.machine.36086: Flags [S.] ...  
14:18:36.786529 IP your.machine.36086 > yummyrecipesforme.com.http: Flags [.] ...  
14:18:36.786589 IP your.machine.36086 > yummyrecipesforme.com.http: Flags [P.] ...  
length 73: HTTP: GET / HTTP/1.1
```



3. Two minutes later — a new DNS lookup for the spoofed domain and a new connection

```
14:20:32.192571 IP your.machine.52444 > dns.google.domain: 21899+ A? greatrecipesforme.com.  
14:20:32.204388 IP dns.google.domain > your.machine.52444: 21899 1/0/0 A 192.0.2.172  
14:25:29.576493 IP your.machine.56378 > greatrecipesforme.com.http: Flags [S] ...
```

ANALYSIS The pivot from `yummyrecipesforme.com` to `greatrecipesforme.com`, on a new source port, is the redirect chain: the downloaded file (disguised as a browser update) sent the victim's browser to a lookalike domain. Combined with the admin lockout, the pattern matches an attacker using a brute-forced admin account to inject malicious redirect code into the original site.

TCP flag codes

Flag	Meaning
[S]	Connection start (SYN)
[S.]	SYN-ACK — server acknowledging the start
[.]	Acknowledgment (ACK)
[P.]	Data push
[F]	Connection finish
[R]	Connection reset

2 Quick Reference

```
tcpdump — quick reference

tcpdump log anatomy:
  timestamp IP source > destination: flags, seq, payload

Flags [S]   connection start      Flags [P.]  data push
Flags [S.]  SYN-ACK response      Flags [F]   connection finish
Flags [.]   acknowledgment       Flags [R]   connection reset

<domain>.http -> port suffix means port 80 (HTTP) on that host
```